A PROJECT SYNOPSIS REPORT

ON

AI-POWERED SMART BANQUET & EVENT PLANNING PLATFORM

In partial fulfillment for the award of the degree of

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1. Introduction

Event planning for occasions such as weddings, engagements, birthdays, receptions, and corporate functions involves several interconnected decisions, including the selection of a suitable venue, the choice of vendors and venues for catering, decoration, photography, and other services, and the management of the overall budget. These decisions become increasingly difficult as the number of guests grows, as the user's requirements change during planning, or when the user has distinct priorities for different services. The task is made more complex by the fact that vendor and venue information, including prices, packages, and availability, changes frequently and is not always reflected accurately across the sources a user consults, which creates a need for a platform in which such information can be dynamically updated by the vendors themselves and in which a plan can be adjusted whenever a customer's requirements change.

The proposed AI-Powered Smart Banquet & Event Planning Platform is a data-driven decision-support system designed to simplify these activities by combining requirement understanding, vendor and venue recommendation, comparison, and budget planning within a single platform. The platform will allow a user to enter event requirements such as event type, guest count, location, budget, and service preferences. Based on these inputs, the system will recommend suitable vendors and venues, compare the available options with a clear explanation of each recommendation, and generate a budget-optimized event plan. When the customer's requirements change after an initial plan has been generated, the system will recalculate the plan and provide updated recommendations and budget allocation through the Event Plan Recalculation feature.

A distinguishing aspect of the platform is that it will be designed with two connected interfaces. On the customer side, users will describe their requirements and receive recommendations, comparisons, and cost estimates. On the vendor side, banquet owners and service vendors will maintain their own prices, packages, services, and date-wise availability through a dedicated dashboard. Because vendors will update this information directly, the data used for recommendation will reflect current prices and availability rather than static listings maintained by a central administrator. Once a booking is confirmed, the agreed price and booking details will be recorded and protected from subsequent changes made to the vendor's listing, and a digital booking

confirmation will be generated for the customer and the vendor, and will be securely maintained by the system.

This synopsis presents the planning and design phase of the project, carried out during the current semester, including the problem definition, literature review, proposed methodology, system architecture, and module design. The complete implementation, integration, and testing of the platform will be carried out in the following semester as part of the Major Project.

1.1 Background and Significance

Traditional event planning requires a user to gather and compare information from multiple independent sources. Venue capacity, price, food options, decoration quality, and photography services are typically evaluated separately, and the user is responsible for combining this information into a coherent plan that fits a defined budget.

The significance of this project lies in three aspects. The first is personalization: two users with an identical budget and guest count may require different recommendations because one may prioritize decoration while another prioritizes food or photography, and a preference-aware scoring approach will represent these priorities and rank available options accordingly. The second is the reliability of the underlying information: because prices, packages, and availability will be maintained directly by vendors rather than by a central administrator, the platform is expected to reduce the likelihood that a user plans around outdated or unavailable options. The third is adaptability: because a customer's requirements can change during planning, the platform will recalculate the recommended plan and budget rather than requiring the process to be repeated manually.

2. Problem Statement

The problem addressed by this project is the absence of an integrated platform that allows a user to select a suitable combination of vendors and venues while receiving a preference-aware recommendation, an optimized cost estimate, and dynamically updated vendor and venue information, including current prices and availability maintained directly by the vendor. A related problem is that a customer's requirements, such as guest count, budget, or event type, can change during the planning process, and existing systems generally do not recalculate the plan automatically when this happens.

Existing event and vendor discovery systems generally present vendors and venues independently rather than as part of a complete event plan, and the information shown is often maintained by a central administrator rather than the vendor, which increases the likelihood of outdated prices or availability. The user is required to manually compare vendors and venues across parameters such as capacity, price, rating, and service quality, and to calculate a complete event budget separately. When a customer's requirements change during planning, the user is required to repeat this comparison and calculation manually in the absence of an integrated recalculation mechanism.

2.1 Gaps in Existing Systems

- Vendor and venue information is handled and presented independently, rather than as part of an integrated event plan.
- Recommendations do not adequately account for individual service priorities.
- Users are required to manually compare parameters such as capacity, price, rating, food quality, decoration, and photography across multiple sources.
- Budget calculation is typically separated from vendor and venue recommendation.
- Vendor and venue information, including price and availability, is often maintained by a central administrator rather than updated directly by the vendor, which increases the likelihood of outdated listings.
- Existing systems generally do not recalculate the event plan when a customer's requirements change.
- A confirmed booking has no standard mechanism to protect its agreed price and details from later changes made to a vendor's listing.

2.2 Need for the Proposed Work

The proposed system will combine requirement understanding, recommendation, comparison, cost optimization, and plan recalculation with vendor-managed data within a single platform. The system will convert user requirements into structured parameters, use these parameters to rank available vendors and venues, and generate a budget-optimized event plan that can be recalculated whenever the customer's requirements change. Vendors will maintain their own prices, packages, and availability directly,

which will keep the information used for recommendation current. The system is intended to support the user's decision-making rather than to replace it or to process real-world financial transactions.

2.3 Research/Development Problem

The development problem is to design a recommendation and cost-optimization pipeline that operates on structured vendor and venue data, incorporates user preferences into a ranking mechanism, produces explainable recommendations, and recalculates the plan when a customer's requirements change. A further requirement is to design a vendor-side data management workflow that keeps vendor and venue information current, together with a booking mechanism that protects the details of a confirmed booking from any subsequent change made to a vendor's listing.

3. Objectives

The main objective of the project is to design an AI-assisted platform that will help users plan events by recommending suitable vendors and venues, comparing available options with clear explanations, generating a budget-optimized event plan that can be recalculated when requirements change, and allowing vendors to independently maintain accurate pricing and availability information.

- To design a structured database containing vendor, venue, service, pricing, availability, and booking information.
- To identify and prepare datasets for vendor- and venue-related data required for recommendation and cost estimation.
- To design a recommendation engine that will rank vendors and venues using a combination of hard filters and preference-based scoring.
- To incorporate user priorities, such as venue, catering, decoration, and photography preferences, into the recommendation scoring mechanism.
- To design a comparison module that will evaluate selected vendors or venues and generate an explanation of the ranking.
- To design a cost estimation and budget optimization module that will propose a combination of vendor and venue services within the user's stated budget.

- To design an event plan recalculation mechanism that will update recommendations and budget allocation when the customer's guest count, budget, event type, or other major requirement changes.
- To design an NLP-based requirement extraction module that will convert natural-language event requirements into structured parameters.
- To design a vendor dashboard that will allow vendors to update prices, packages, services, and date-wise availability, including offline bookings.
- To design a booking confirmation mechanism that will protect the price and details of a confirmed booking from subsequent listing changes, and generate a digital booking confirmation record.
- To plan the evaluation approach for recommendation relevance, cost-calculation correctness, and overall functional performance, to be carried out during implementation.

3.1 Key Functional Goals

Goal	Expected Function
Personalization	Rank vendors and venues according to the user's stated priorities.
Budget optimization	Recommend a combination of vendor and venue services that fits the stated budget.
Explainability	Present the factors that support each recommendation.
Adaptability	Recalculate the plan and budget when the customer's requirements change.
Vendor-managed data	Allow vendors to maintain current prices, packages, and availability.
Booking protection	Preserve the agreed price and details of a confirmed booking.

4. Scope of the Project

The scope of the project covers the planning, design, and prototype implementation of a platform that will demonstrate preference-aware vendor and venue recommendation, budget optimization, event plan recalculation, vendor-managed pricing and availability, and protected booking confirmation.

4.1 Included in Scope

- Vendor and venue discovery and filtering based on event type, guest count, location, and budget.
- Preference-based ranking and comparison of vendors and venues for services such as catering, decoration, photography, and DJ/entertainment.
- An explanation of the factors supporting each recommendation.
- Cost estimation and budget optimization across vendor and venue service categories.
- Recalculation of the recommended plan and budget when the customer's guest count, budget, event type, or other major requirement changes.
- NLP-based extraction of event requirements from natural-language input.
- A vendor/banquet dashboard for managing prices, packages, services, and date-wise availability, including offline bookings.
- Booking confirmation with protection of the agreed price and details after confirmation, and generation of a digital booking confirmation record.
- MySQL-based storage of vendor, venue, pricing, availability, and booking information.
- Python-based data processing and machine learning.
- A Streamlit-based application interface for both the customer and vendor dashboards.

4.2 Constraints

- Real-world payment processing is not a mandatory part of the prototype; the booking confirmation will record an agreed amount without processing an actual financial transaction.
- The accuracy of recommendations and cost estimates will depend on the quality and coverage of the dataset used.
- A dispute concerning an already confirmed booking will be resolved directly between the customer and the vendor and remains outside the scope of the platform.

- The recommendations generated by the system are intended as decision support and should be verified by the user before a final commitment is made.

4.3 Current Semester Scope and Next Semester Plan

The work presented in this synopsis corresponds to the planning and design phase of the project, carried out during the current semester. This phase covers problem identification, requirement analysis, literature review, system planning, dataset identification and preparation, selection of appropriate AI/ML approaches, architecture and database design, and module planning. The complete implementation, coding, integration, and testing of the proposed system will be carried out in the following semester as part of the Major Project.

5. Literature Review

The proposed platform draws on concepts from recommender systems, machine learning, natural language processing, optimization, and two-sided marketplace design. The following subsections summarize the relevant background and identify the research gap addressed by the proposed platform.

5.1 Recommender Systems

Recommender systems assist users in identifying items relevant to their preferences and are commonly implemented using collaborative filtering, content-based recommendation, or hybrid approaches (Ricci, Rokach, and Shapira, 2022). Collaborative filtering depends on interaction data collected from a large number of users, which is difficult to obtain within the scope of a student prototype with a limited initial user base. A content-based or attribute-based approach is therefore more appropriate for this project, since vendor and venue records can be represented using structured features such as price, capacity, location, and service ratings, and compared using similarity or weighted-scoring techniques (Resnick and Varian, 1997). The proposed platform will adopt this content-based direction rather than a collaborative approach.

5.2 Machine Learning for Ranking and Optimization

Machine learning techniques convert item and user attributes into a relevance score after appropriate feature scaling and categorical encoding (Pedregosa et al., 2011).

Beyond ranking individual options, selecting a complete vendor-and-venue combination within a fixed budget is a constrained combination-selection problem, related to classical optimization techniques such as knapsack-style allocation, in which the objective is to maximize preference satisfaction subject to a budget constraint (Geron, 2022). The proposed platform will apply scoring-based ranking for recommendation and a constrained selection approach for budget optimization, rather than relying on either technique alone.

5.3 Natural Language Processing for Requirement Extraction

Natural language processing allows a user to describe event requirements in free-form text rather than through a fully structured form. Rule-based pattern matching and named-entity recognition techniques, commonly available in NLP libraries, are used to extract structured slots such as guest count, budget, location, and event type from short natural-language statements. For a project of this scale, a rule-based or library-based extraction approach is more realistic than training a language model on a large corpus, and will be adopted in the proposed methodology.

5.4 Data-Driven Cost Estimation

Cost estimation for a multi-component service, such as an event, can be implemented through transparent, component-level formulas that combine guest-dependent and package-dependent costs. More advanced predictive estimation, using regression techniques trained on historical pricing data, can be introduced once sufficient data becomes available. Given the data realistically obtainable within a student project, a transparent, formula-based approach will be adopted as the primary cost-estimation method, with the option to extend it using regression-based prediction as the dataset grows.

5.5 Vendor-Managed Marketplace Platforms

Two-sided platforms, in which the supplier of a service maintains their own listing rather than relying on a central administrator, are a well-established design pattern in online marketplaces. Allowing the vendor to directly control price, package, and availability information keeps the underlying dataset current and reduces the administrative effort otherwise required to maintain the platform. This principle directly informs the design of the proposed vendor dashboard.

5.6 Research Gap and Contribution

The systems and techniques reviewed above are typically applied in isolation: recommendation systems rank items, optimization techniques allocate constrained resources, and marketplace platforms allow supplier-managed data, but existing banquet and event-planning platforms rarely combine all of these within a single, connected workflow. Most existing platforms present vendors and venues as independent, centrally maintained listings, without a preference-aware ranking mechanism, an integrated budget-optimization step, or a mechanism to recalculate the plan when a customer's requirements change. This gap forms the basis for the proposed platform, which will combine natural-language requirement extraction, preference-aware recommendation with explanation, budget optimization, event plan recalculation, and vendor-managed pricing and availability within a single, connected system.

6. Methodology / Proposed System

The proposed system will follow a modular pipeline with two connected interfaces. On the customer side, requirements will be collected through a structured form or natural-language input and converted into structured parameters that will be passed to the recommendation and budget optimization components. On the vendor side, a separate dashboard will allow vendors to maintain prices, packages, services, and availability, which will directly update the data used by the customer-side modules. This section presents the proposed methodology to be implemented during the following semester.

6.1 Proposed Architecture

The application layer will be implemented using Streamlit, with separate views for the customer dashboard and the vendor dashboard. Python will provide the application logic, data processing, and machine learning functionality. Pandas and NumPy will be used for data manipulation and numerical operations, and MySQL will store structured vendor, venue, pricing, availability, and booking information. An NLP layer will support requirement extraction on the customer side.

Conceptual flow: Customer Input (Form/NLP) → Requirement Extraction → Recommendation and Comparison Engine → Budget Optimization → Event Plan Recalculation (on requirement change) → MySQL (Vendor, Venue, and Booking Data)

↔ Vendor Dashboard (Price and Availability Updates) → Booking Confirmation Module → Digital Booking Confirmation.

6.2 Overall Workflow

Stage	Description
Customer input	Event type, location, guest count, budget, and service preferences.
Requirement extraction	Convert form or natural-language input into structured parameters.
Data retrieval	Retrieve current vendor and venue records from MySQL.
Filtering	Remove options that do not satisfy essential constraints, such as capacity or location.
Recommendation and comparison	Calculate relevance scores, rank suitable options, and generate an explanation.
Budget optimization	Select a combination of vendor and venue services that fits the stated budget.
Requirement change	Customer updates guest count, budget, event type, or another major requirement.
Plan recalculation	Recommendation and budget optimization modules are re-run to generate an updated plan.
Vendor update	Vendor modifies price, package, service, or availability information through the vendor dashboard.
Booking confirmation	Record the agreed price and details, and generate a digital booking confirmation.
Output	Recommended plan, estimated budget, and booking confirmation record.

6.3 Algorithms and Processing Logic

The system will use a combination of deterministic business rules and similarity- or scoring-based ranking techniques. This hybrid approach is appropriate because some requirements, such as guest capacity, are strict constraints, while others, such as service priorities, are preferences that should influence ranking rather than eliminate options.

Algorithm A: Preference-Based Ranking

Each candidate vendor or venue will be assigned a normalized score based on multiple weighted features:

$$\text{Score} = w_V(\text{Venue}) + w_F(\text{Food}) + w_D(\text{Decoration}) + w_P(\text{Photography}) + w_B(\text{Budget Fit}) + w_L(\text{Location})$$

The weights will be set according to the priorities stated by the user during requirement collection; for example, if decoration is marked as a high priority and photography as a low priority, the decoration weight will be increased and the photography weight reduced. The exact weight values will be calibrated and validated during implementation.

Algorithm B: Budget Optimization

Budget optimization will be treated as a constrained combination-selection problem rather than a simple additive calculation. Given the ranked vendors and venues for each service category, the module will select the combination that best satisfies the user's preferences while keeping the total estimated cost within, or close to, the stated budget:

$$\text{Total Cost} = \text{Venue Cost} + (\text{Guests} \times \text{Food/Plate Cost}) + \text{Decoration Cost} + \text{Photography Cost} + \text{Other Service Costs}$$

When the estimated cost of the best-ranked combination exceeds the stated budget, the module will propose lower-cost alternatives for one or more service categories and recalculate the total. When the customer's requirements change after a plan has already been generated, this recalculation will be performed through the Event Plan Recalculation feature described in Section 6.4.

6.4 Event Plan Recalculation Logic

Event Plan Recalculation is a core feature of the proposed platform. When a customer changes a major requirement after an initial plan has been generated — such as the guest count, budget, event type, or a selected service — the system will re-run the recommendation and budget optimization modules using the updated parameters and will present an updated set of recommendations and budget allocation. This allows the customer to explore the effect of a changed requirement without needing to restart the planning process from the beginning. The recalculation logic will reuse the same scoring and optimization methods described in Section 6.3, applied to the updated requirement values.

6.5 Vendor-Side Data and Availability Management

Price, package, and availability management will be implemented as direct database update functionality rather than a machine-learning component: a vendor will update a

record through the vendor dashboard, and the corresponding value will be immediately reflected in the data used by the recommendation and budget optimization modules. This distinction will be maintained clearly in the system design, since maintaining current listings is a data-management function and not an artificial intelligence function.

When a vendor receives an offline booking for a given date, the vendor will mark that date as unavailable through the dashboard. The platform will then exclude that date from the availability shown to other customers for that vendor or venue, which will reduce the likelihood of a scheduling conflict between an offline booking and a booking made through the platform.

6.6 Booking Confirmation and Data Protection Logic

When a customer confirms a booking, the system will store the agreed price and booking details as a fixed record associated with that specific booking, separate from the vendor's current price and availability record. Any subsequent change made by the vendor to the current price or availability will update the vendor's listing for future customers only and will not alter the details of an existing confirmed booking. A dispute concerning an already confirmed booking will be handled directly between the customer and the vendor and will not be automatically resolved by the platform.

A digital booking confirmation record will be generated at the time of confirmation and will contain the customer name, vendor or venue name, event date, selected package or services, the agreed booking amount, a unique booking confirmation identifier, the booking status, and the confirmation date. This record will be made available to the customer and the vendor, and will be securely maintained by the system.

6.7 Tools and Technologies Used

Technology/Tool	Purpose in Project
Python	Core programming, application logic, and AI/ML implementation.
Pandas	Data cleaning, transformation, and analysis.
NumPy	Numerical operations and feature processing.
Scikit-learn	Preprocessing and similarity/ranking computations.

Technology/Tool	Purpose in Project
Streamlit	Web-based interface for the customer and vendor dashboards.
MySQL	Storage of vendor, venue, pricing, availability, and booking data.
MySQL Connector / SQLAlchemy	Database connectivity from Python.
NLP library (e.g., spaCy/rule-based)	Extraction of structured parameters from natural-language input.
Matplotlib / Plotly	Visualization of budget breakdown and comparison results.
Git/GitHub	Version control and collaborative development.

6.8 Proposed Data Relationships

A user will be able to create multiple events, and each event will be associated with one requirement profile and one or more recommendation results. A venue will be associated with a vendor profile and can offer multiple packages, and a vendor can provide services across multiple venues or independently, such as catering, decoration, or photography. Each confirmed booking will be associated with exactly one event, one vendor-venue combination, and one digital booking confirmation record. These relationships will be represented using primary and foreign keys in the proposed MySQL schema.

7. Module Description

The proposed system is planned to include the following modules, to be implemented during the following semester.

7.1 Data Collection Module

This module will maintain vendor, venue, service, and pricing records. The initial prototype dataset will be created from structured sample data and, where permitted, publicly available information. Each record will contain consistent fields to support recommendation and cost estimation.

7.2 Data Preprocessing Module

Pandas will be used for cleaning, transformation, and validation of the dataset. Missing values, inconsistent text values, duplicate records, and numerical formatting issues will

be handled before the data is used by the recommendation module. Categorical attributes will be encoded and numerical features normalized where required.

7.3 Event Requirement Understanding Module (NLP)

This module will convert a customer's natural-language or form-based input into structured parameters such as event type, guest count, budget, location, and service preferences, using rule-based pattern matching and NLP techniques appropriate to the scale of the project.

7.4 Recommendation and Comparison Module

This module will combine hard filters, such as capacity and location, with preference-aware scoring to produce a ranked list of suitable vendors and venues. The module will return a ranked list together with an explanation of the leading factors behind each recommendation, rather than a single unexplained result.

7.5 Budget Optimization Module

This module will estimate the cost of the recommended vendor and venue combination and will propose adjustments when the estimated cost exceeds the user's stated budget.

7.6 Event Plan Recalculation Module

This module will re-run the recommendation and budget optimization modules whenever the customer changes a major requirement, such as guest count, budget, or event type, and will present an updated plan without requiring the customer to restart the process.

7.7 Vendor Dashboard Module

This module will provide a separate interface through which a vendor can update prices, packages, services, and date-wise availability, and mark dates as unavailable following an offline booking. This module performs standard data-management operations and will not include a machine-learning component.

7.8 Booking and Confirmation Module

This module will record a confirmed booking as a fixed entry containing the agreed price and details, independent of subsequent changes to the vendor's current listing, and will generate the digital booking confirmation record described in Section 6.6.

7.9 Database Module

MySQL will store users, vendors, venues, services, pricing, availability, requirement profiles, and booking and confirmation records, and will support the data retrieval required by the recommendation, budget optimization, and vendor dashboard modules.

7.10 Implementation (Planned)

The proposed system will be implemented by integrating the recommendation and comparison module, budget optimization module, event plan recalculation module, NLP-based requirement extraction, vendor dashboard, booking confirmation module, and the MySQL database within a Streamlit application. This implementation will be carried out during the following semester.

7.11 Testing (Planned)

The system will be tested to verify the correctness and reliability of each module once implementation is complete. Planned test cases will cover requirement extraction accuracy, recommendation and comparison output, budget calculation correctness, event plan recalculation, vendor-side update propagation, booking confirmation and price protection, database operations, and overall system performance.

8. Expected Outcomes

The expected outcome of the completed project is a working prototype that will demonstrate how preference-aware recommendation, budget optimization, plan recalculation, and vendor-managed data can support event-planning decisions. The outcomes listed below are planned deliverables of the following semester's implementation phase.

- A user-friendly interface for entering event requirements through a structured form or natural language.
- Personalized vendor and venue recommendations based on budget, capacity, location, and preferences.
- Vendor recommendations for services such as catering, decoration, and photography, aligned with user priorities.
- A clear explanation of the leading factors behind each recommendation.

- An approximate, budget-optimized event cost with a component-wise breakdown.
- Automatic recalculation of the recommended plan and budget when the customer's requirements change.
- A vendor dashboard through which vendors independently maintain prices, packages, services, and availability.
- A protected, fixed record of the agreed price and details for each confirmed booking.
- A digital booking confirmation record accessible to the customer and the vendor.
- Structured storage of vendor, venue, pricing, availability, and booking data in MySQL.

8.1 Advantages

- Personalized recommendations in place of generic, unranked listings.
- Integrated recommendation, comparison, budget optimization, and plan recalculation within a single workflow.
- Vendor-managed pricing and availability, which will keep the underlying dataset current.
- A protected booking record that will keep a confirmed booking unaffected by later vendor-side changes.
- A modular architecture that can be extended with additional service categories.

8.2 Evaluation (Planned)

Once implemented, the prototype will be evaluated using functional test cases, recommendation-relevance checks against manually curated test scenarios, cost-calculation test cases, and usability feedback. Where a supervised machine-learning model is used, standard metrics such as precision, recall, F1-score, or ranking metrics such as Precision@k will be reported according to the model selected.

8.3 Limitations

- The accuracy of recommendations and cost estimates will depend on the quality and coverage of the dataset used.
- Prototype prices may differ from real, negotiated prices at the time of an actual booking.
- Availability information will reflect the updates maintained by vendors through the dashboard rather than an external, real-time source.
- The performance of more advanced machine-learning models will depend on the size and quality of the dataset available at the time of implementation.

8.4 Future Scope

- Real-time payment processing and confirmed online transactions.
- Map-based location and travel-time consideration for vendor and venue selection.
- Dynamic pricing informed by historical booking data.
- More advanced hybrid recommendation models as sufficient interaction data becomes available.
- Multilingual interaction, including Hindi/Hinglish input support.
- Calendar integration and automated reminders for confirmed events.
- Cloud deployment and a scalable multi-user architecture.

8.5 Innovation Potential

The strongest innovation opportunity lies in combining preference-aware recommendation and explanation, budget optimization, automatic plan recalculation, and a vendor-managed two-sided data model within a single platform. Rather than functioning as a static listing service, the platform is designed to maintain current information through direct vendor participation, to adapt the plan when a customer's requirements change, and to protect the integrity of a confirmed booking, moving the system from a simple search interface toward an integrated planning and booking platform.

9. Hardware/Software Requirements

9.1 Hardware Requirements

Component	Minimum/Recommended
Processor	Intel Core i3 or equivalent; i5 or equivalent recommended.
RAM	8 GB minimum; 16 GB recommended.
Storage	At least 10–20 GB of free project space.
Display	Standard laptop or PC display.
Internet	Required for external libraries, package installation, and any external NLP resources.

9.2 Software Requirements

- Operating System: Windows 10/11 or equivalent.
- Python 3.x.
- VS Code or an equivalent development environment.
- MySQL Server and MySQL Workbench.
- Pandas, NumPy, and Scikit-learn.
- Streamlit.
- MySQL Connector or SQLAlchemy.
- An NLP library for requirement extraction.
- Git/GitHub for version control.

9.3 Data Requirements

The system will require structured data for vendors, venues, capacities, prices, ratings, service attributes, and package costs, together with availability status for each vendor or venue. The dataset will be documented with field definitions and a clearly stated source or generation method.

10. Work Plan / Timeline (Current Semester)

The project is currently at the planning and design stage. The following six-week plan covers the remaining work of the current semester, consisting of documentation, research, analysis, dataset identification, and system planning. The complete design

implementation, coding, integration, and testing of the proposed system will be carried out in the following semester as part of the Major Project.

Week	Planned Work	Expected Deliverable
1	Problem identification and requirement analysis.	Finalized problem statement and objectives.
2	Literature review and study of existing systems.	Completed literature review.
3	System architecture and module planning.	Proposed architecture and module list.
4	Dataset identification, preparation, and database planning.	Identified data sources and draft database schema.
5	Selection of AI/ML approaches and methodology preparation.	Finalized methodology and algorithms.
6	Documentation, review, and synopsis finalization.	Completed and submitted synopsis.

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12. Conclusion

The AI-Powered Smart Banquet & Event Planning Platform is proposed as a data-driven decision-support system for personalized, budget-aware, and adaptable event planning. During the current semester, project work has focused on problem identification, requirement analysis, literature review, system architecture, database and module planning, and preparation of the proposed methodology, as presented in this synopsis.

The proposed system is designed to represent what the customer actually values: a user may prefer a premium venue and decoration while accepting a standard photography package, and the recommendation engine is designed to represent these priorities rather than treating every service equally. The budget optimization module will complement this recommendation by proposing a combination of vendor and venue services that fits the stated budget, and the Event Plan Recalculation feature will update this plan whenever the customer's requirements change, without requiring the process to be repeated manually.

The vendor dashboard is designed to allow prices, packages, services, and availability to be maintained directly by the vendor, which will keep the information used for recommendation current, while the booking confirmation mechanism is designed to ensure that the price and details of a confirmed booking remain protected from any later change to a vendor's listing. The proposed implementation, using Python, Pandas, Scikit-learn, Streamlit, and MySQL, is intended to provide a practical and manageable technology stack for a B.Tech final-year project.

The complete implementation, integration, and testing of the proposed system will be carried out in the following semester as part of the Major Project. The overall success of the project will depend on the quality of the dataset prepared, the design of the recommendation and optimization logic, the correctness of the cost calculations, and the reliability of the vendor-side data management and booking confirmation mechanisms.